NUS SCHOOL OF COMPUTING · CS5224 CLOUD COMPUTING · TEAM 18

SeatRelay

One waitlist for every session of a workshop.

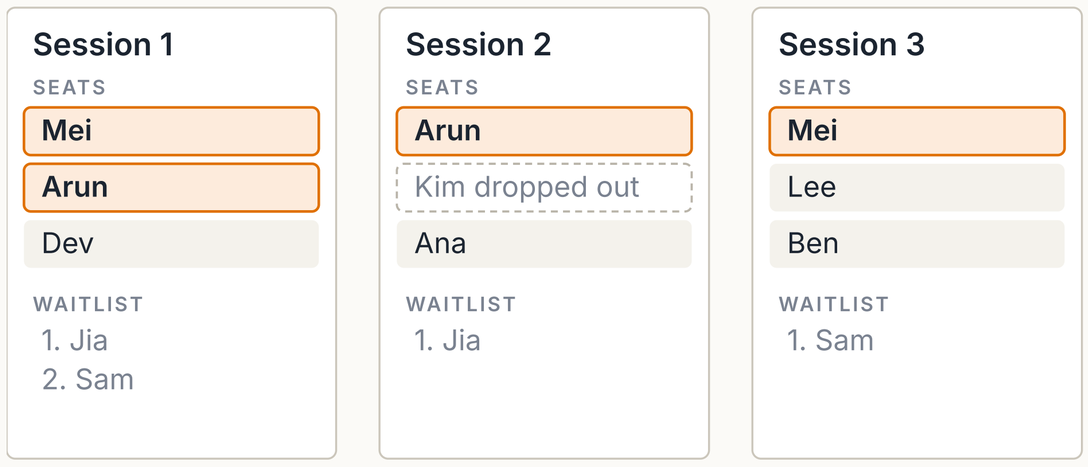
Register once and rank the sessions you can attend. When a seat frees up, it is held for the next person in line automatically, instead of staying empty or waiting for an organiser's e-mail.

Li Zongze · Chen Yang · Lin Jie · Zhu Gengyu · Liang Zihan · Zhang Tianyi

Try the live demo
d2ni5e1z674t04.cloudfront.net

1 Why one form per session wastes seats

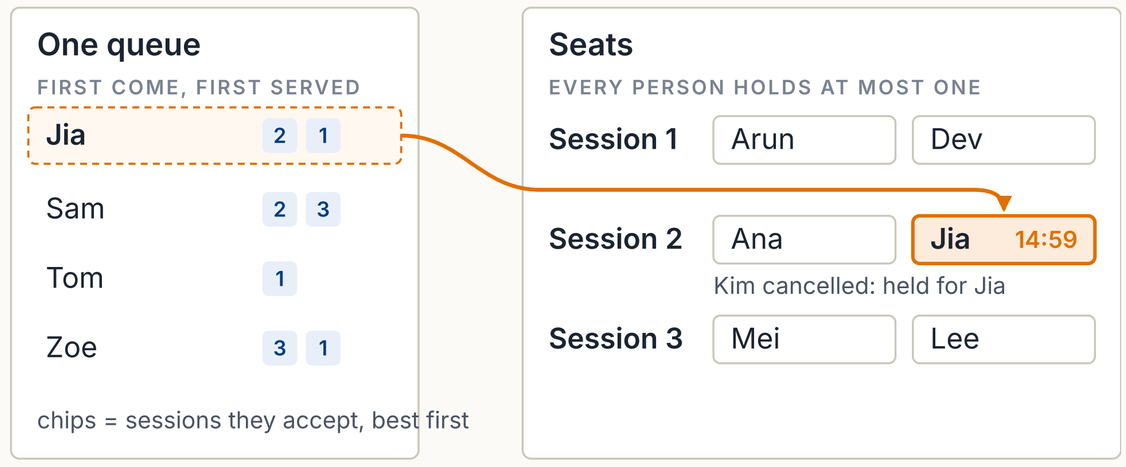
Today: a separate sign-up form for each session



Mei and Arun each hold two seats to be safe; someone else misses out.

Kim's seat stays empty until an organiser e-mails Jia by hand.

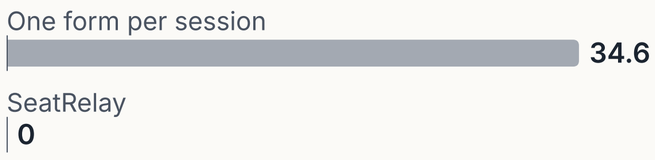
SeatRelay: one queue, at most one seat each



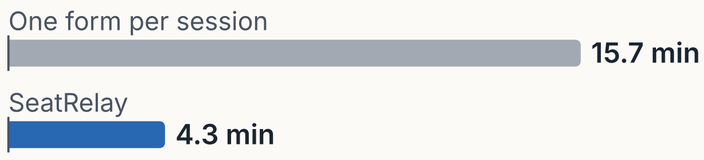
A freed seat is held for the first person who accepts that session, with a countdown.

No organiser step: the hand-over is part of the same database decision.

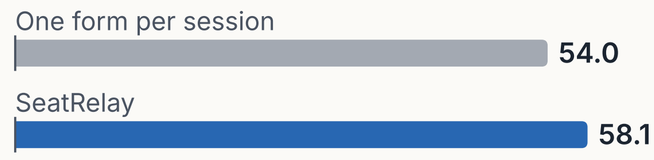
Duplicate holds per workshop



Minutes until a freed seat is filled



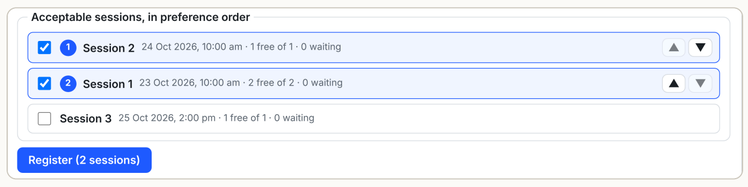
Different people who got a seat



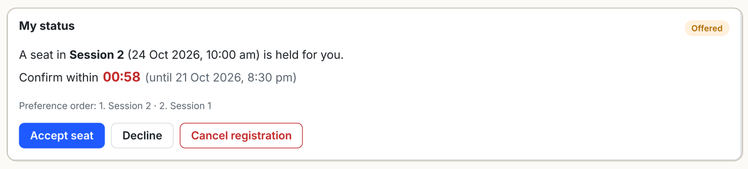
Simulation, not field data: our allocation engine on 1,000 random workshops, the same arrivals replayed through both designs, with assumed participant behaviour (mean per workshop).

2 How it works

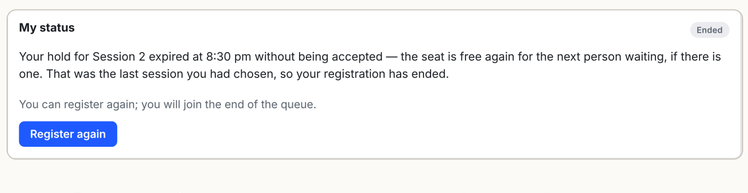
1 Tick every session you could attend and put them in order.



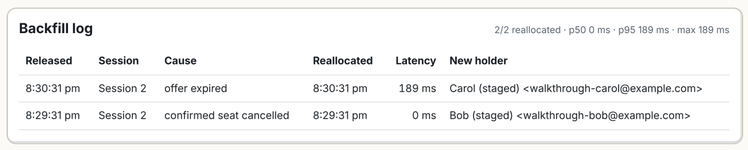
2 Wait in one queue. When a seat in one of your sessions frees up, it is held for you (15 min by default).



3 Accept, decline or let it expire: the seat moves on to the next person at once.



4 Organisers see every hand-over and its delay.

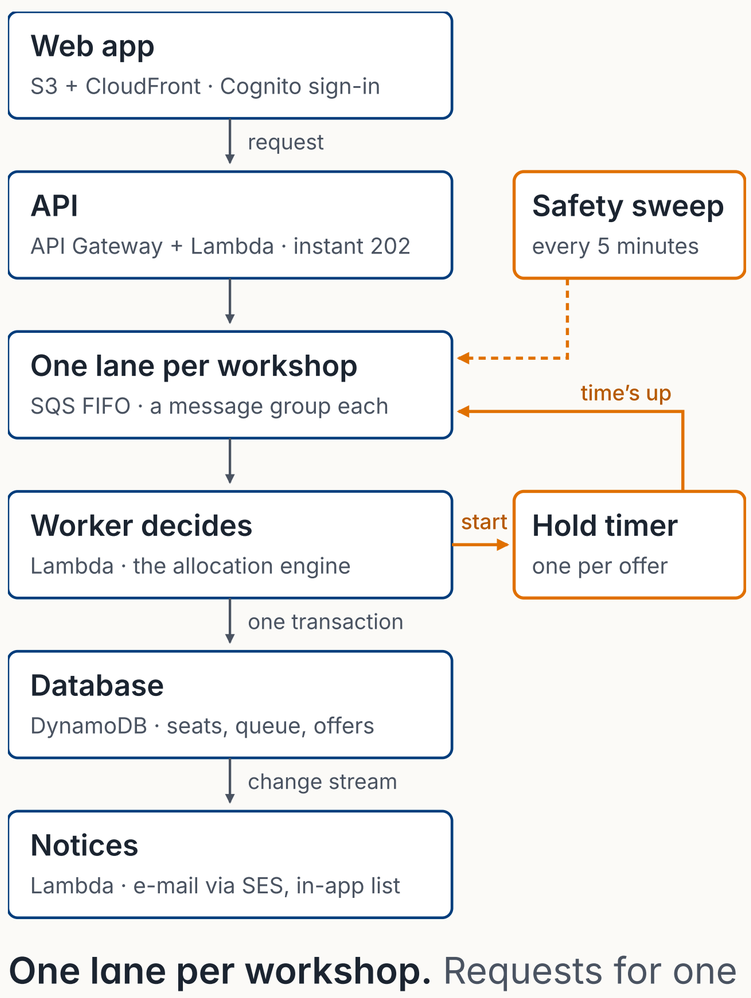


Backfill log 2/2 reallocated · p50 0 ms · p95 189 ms · max 189 ms

Released	Session	Cause	Reallocated	Latency	New holder
8:30:31 pm	Session 2	offer expired	8:30:31 pm	189 ms	Carol (staged) <walkthrough-carol@example.com>
8:29:31 pm	Session 2	confirmed seat cancelled	8:29:31 pm	0 ms	Bob (staged) <walkthrough-bob@example.com>

Screens of the deployed app in a staged run with test accounts (1-minute holds).

3 What runs on AWS



One lane per workshop. Requests for one workshop are decided one at a time, so they never race; different workshops run in parallel.

All or nothing. Seats, queue, offer and notice change in one database transaction.

A timer for every hold. Expiry is judged by server time, so a late or repeated timer cannot touch a newer offer.

4 Measured on AWS

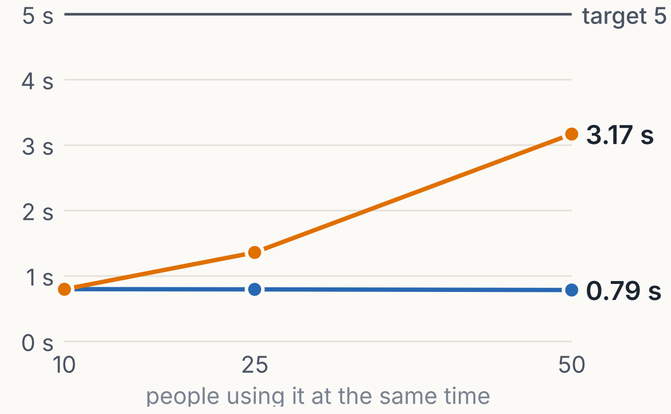
8/8 acceptance tests from our project proposal pass on the deployed system

0 seats overbooked or held twice in 672 checks, with 50 people racing for 5 seats

0.10 s for a freed seat to reach the next person (95th percentile of 300; target 120 s)

2 min to recover from a server crash mid-update, applied exactly once (target 5 min)

Time from request to final answer, as the user sees it (95th percentile)



k6 load test, 3 rounds per level, 0.00 % errors. One workshop is decided one request at a time, which is why it slows first.

US$13.30 → US$5.91 per 1,000 registrations as volume grows from 1,000 to 100,000 a month (AWS list price, our cost model).

Next: a user study with twelve students and a live two-session pilot with real participants.

Code, data and every measurement: github.com/gengyuzccc-cell/Team18_CS5224_SeatRelay_Project_AY2627Sem1 (results/). Measured on our AWS deployment (ap-southeast-1) on 17 Oct 2026. **AI use:** an AI coding assistant (Claude Code) helped write and test the code, run the experiments and lay out this poster; the requirements and design decisions are the team's, and every number here is generated from the measurement files in the repository.

code & data